

Analytic Epidemiology - Study Designs I
APHS 20203 • Epidemiology & Biostatistics

Epidemiological Study: Basic Question
Are exposure and outcome associated?
More specifically:

- Do people with different exposures experience different health outcomes?
- Is the observed association likely to be causal, or could it reflect other explanations?

Classification of Study Designs

Two Categories of Analytic Studies

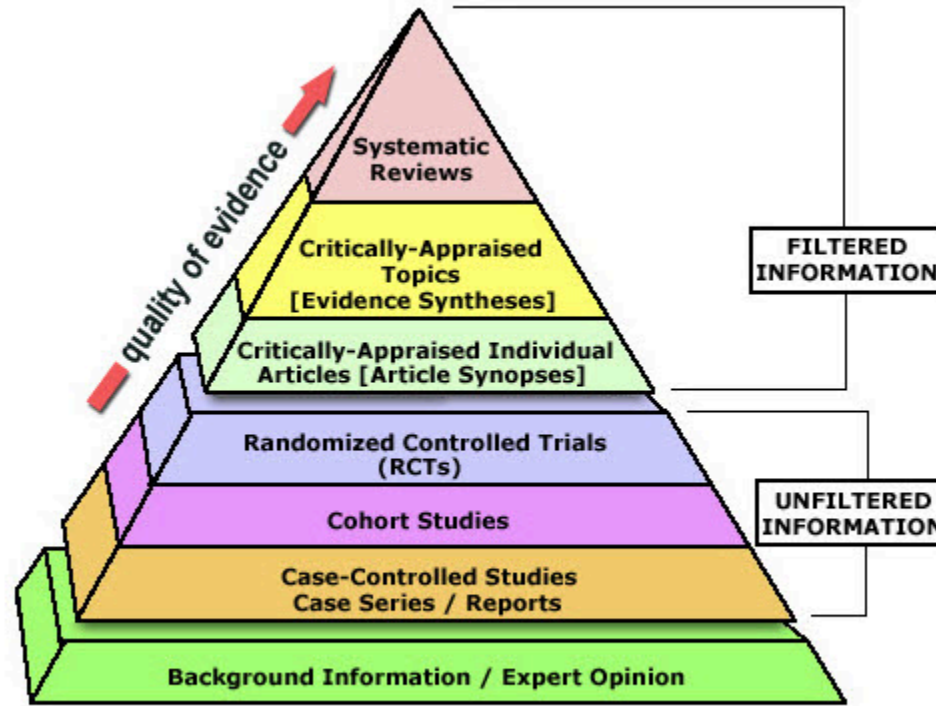
Observational design

- Investigator does not assign exposure
- Observes existing differences in exposure and outcome
- Primary goal: identify associations
- Examples include ecologic, cohort, and case-control studies

Experimental design

- Investigator assigns exposure or intervention
- Often uses randomization
- Primary goal: evaluate causal effects
- Examples include randomized controlled trials (RCTs)

Levels of Evidence in Epidemiologic Research



Generate Hypotheses
Establish Causality

Study designs vary in how strongly they support causal inference

Lower levels are often used to generate hypotheses

Higher levels are better suited to test causal effects

Moving up the pyramid generally means:

Greater control over alternative explanations

Stronger ability to evaluate causality

Important note:

Higher evidence does not mean “always correct,” and lower evidence does not mean “wrong.”

Epidemiologic Inference: From Association to Conclusion

Analytic epidemiology asks whether an observed association reflects a causal relationship.

Step 1: Identify an association: Do exposure and outcome occur together more often than expected?

Step 2: Evaluate validity:

Internal validity: Is the association likely to be real within this study?

External validity: Can the findings be generalized to other populations?

Step 3: Consider alternative explanations

Could the association reflect chance?

Could it reflect systematic differences in how groups are compared or observed?

Could it reflect other factors related to both exposure and outcome?

Key takeaway: Study design and validity assessment determine how confidently we can move from association to inference.

Ecologic Studies: Limits of Causal Inference

Ecologic Studies: Group-level analysis

Unit of analysis: groups (not individuals)

Exposure & outcome: measured at the population level

Common use: explore population patterns; generate hypotheses

Key caution: group-level associations $\neq$ individual-level conclusions

Take-home message: Ecologic studies are useful for population questions, but cannot establish individual-level causality.

Ecologic Fallacy

Definition: Incorrectly applying group-level findings to individuals

Occurs when associations observed in populations are assumed to hold at the individual level

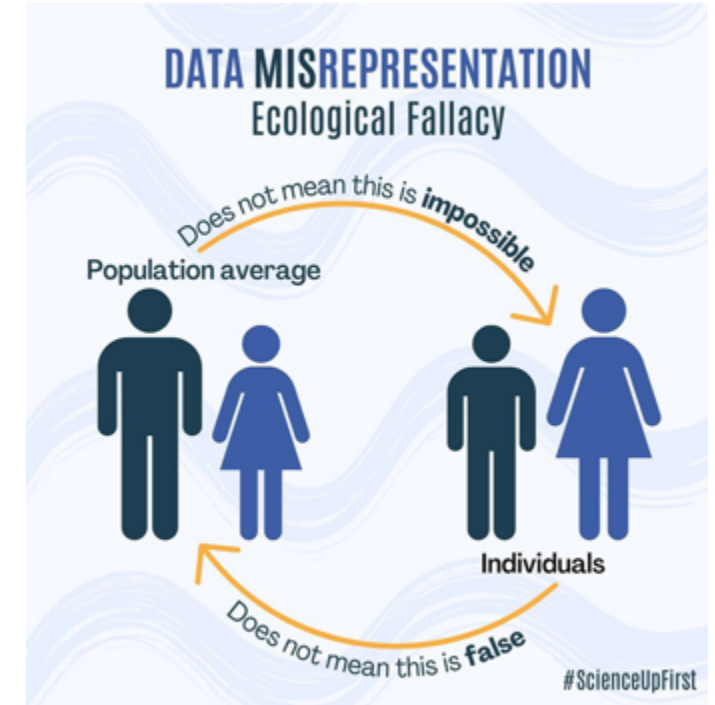
Example

Observation: Cities with higher income have higher CHD rates

Incorrect inference: Wealthier individuals have higher CHD risk

Reality: Within cities, CHD risk may be higher among lower-income individuals

Take-home message: Group-level data cannot be used to draw individual-level conclusions.



<https://www.scienceupfirst.com/project/data-misrepresentation-ecological-fallacy/>

Ecologic Study: Strengths & Limitations

Strengths

- Uses existing population-level data
- Efficient: fast and low cost
- Useful when individual-level data are unavailable

Limitations

- Ecologic fallacy
- Crude exposure measurement
- Limited control of confounding

Discussion: Ecologic Study Example

Study

Sasaki et al. (1993): dietary fat intake & breast cancer mortality

Unit of analysis: countries (30 countries total)

Key finding

Higher animal fat intake was associated with higher breast cancer mortality

Fish fat intake was associated with lower breast cancer mortality

Discussion prompts

What makes this study feasible and efficient?

Why can these results not be interpreted at the individual level?

What other population-level factors could explain the association?

Sasaki S, Horacsek M, Kesteloot H. An ecological study of the relationship between dietary fat intake and breast cancer mortality. *Prev Med.* 1993 Mar;22(2):187-202. doi: 10.1006/pmed.1993.1016. PMID: 8483858.

Experimental Studies

- Key feature: investigator assigns exposure or intervention

- Assigning exposure allows stronger conclusions about cause and effect.

- Comparison: intervention group vs control group

- Goal: evaluate the effect of an intervention

- Two broad types

- True experimental studies: include randomization

- Quasi-experimental studies: no randomization, but intervention is assigned

- Why they matter

- Stronger control of alternative explanations

- Better support for causal inference than observational designs

Intervention Studies

Key Features

- A type of experimental study

- Introduce a planned action to assess its impact

- Conducted in real-world settings

Common examples

- Medical interventions (e.g., vaccines, treatments)

- Behavioral programs (e.g., smoking cessation)

- Policy or community interventions (e.g., taxation, education)

- Take-home message: Intervention studies evaluate whether deliberate actions change health outcomes in practice.

Experimental vs Intervention Studies

Experimental studies

- Broad category where the investigator manipulates a condition

- Often conducted in controlled settings

- Example: laboratory study testing how temperature affects a biological process

Intervention studies

- A type of experimental study focused on planned actions to improve health

- Conducted in real-world populations

- Example: introducing a vaccination program or a smoking cessation policy

- Key point: All intervention studies are experimental, but not all experimental studies are health interventions.

Exercise: Experimental vs Intervention Studies

Exercise: For each example below, decide whether it is an Experimental study or an Intervention study.

Researchers expose bacteria to different temperatures in a lab and measure growth rate.

A city introduces a new bike-lane policy to encourage physical activity and reduce injuries.

Participants are randomly assigned to receive a new medication or a placebo to evaluate treatment effectiveness.

Psychologists test whether blue or red lighting affects concentration in a controlled setting.

Hint

Which examples involve a planned action to improve health?

Which examples involve manipulation without a health intervention?

Clinical Trials: Medical Intervention Studies

Clinical Trials

A type of intervention study conducted in humans

Participants receive a medical intervention under controlled conditions

Designed to evaluate safety and effectiveness

Common purposes

Prevention (prophylactic trials, e.g., vaccines)

Treatment (therapeutic trials, e.g., new drugs or procedures)

Key takeaway: Clinical trials provide strong evidence for whether medical interventions work and are safe.

- Randomized Controlled Trials (RCTs)
 - Randomized Controlled Trials (RCTs)
 - A type of clinical trial
 - Participants are randomly assigned to intervention or control groups
 - Groups are comparable at baseline by design
 - Outcomes are measured consistently across groups
 - Why randomization matters
 - Reduces systematic differences between groups
 - Strengthens causal inference
- Take-home message: Randomization helps isolate the effect of the intervention, making RCTs the strongest design for evaluating causality.

- Randomization
 - Assigns participants to groups by chance
 - Balances known and unknown factors across groups
 - Occurs before outcomes are measured
- What randomization does
 - Reduces systematic differences between groups
 - Minimizes alternative explanations for observed effects

RCT: Randomization

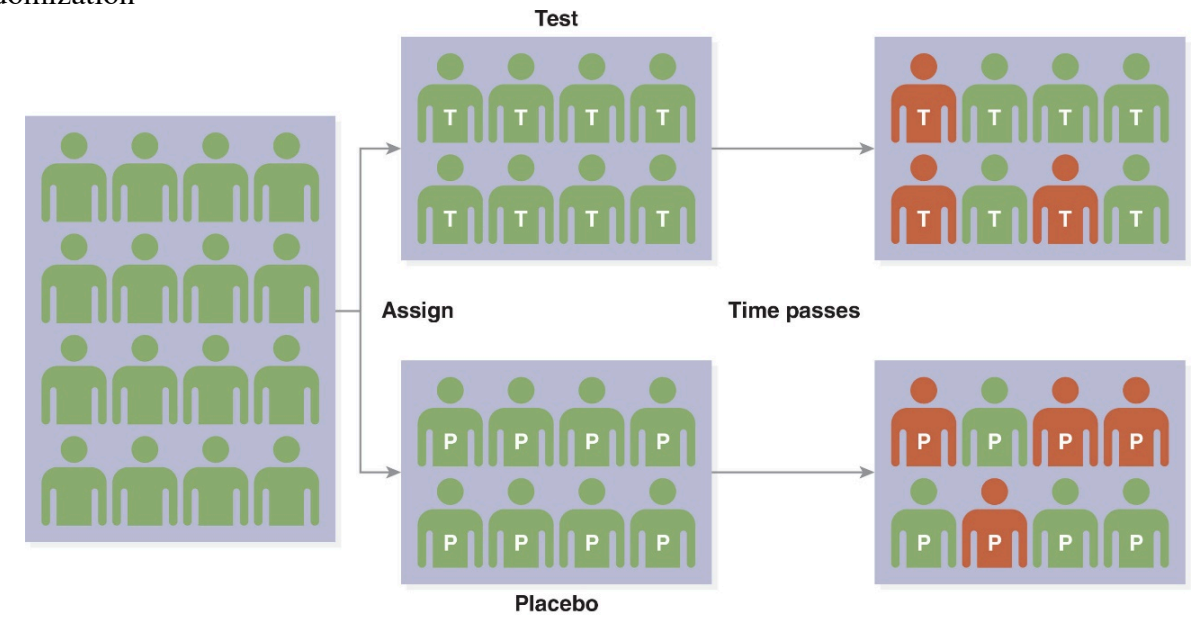


Figure 7.5 Diagram of a randomized controlled trial

Indication that red means occurrence of the outcome.

Take-home message

Randomization improves internal validity by making groups comparable at baseline.

Discussion: Randomized Controlled Trial (RCT) Example

Study overview

- Large randomized controlled trial of a low-fat dietary pattern
- Postmenopausal women assigned to intervention or usual diet

Long-term follow-up

Discussion prompts

- Why is an RCT an appropriate design for this question?

- What features of this study strengthen causal inference?

- What challenges might arise in long-term behavioral interventions?

- Key takeaway: RCTs strengthen causal conclusions, but feasibility, adherence, and follow-up matter.

Prentice RL, Caan B, Chlebowski RT, et al. Low-Fat Dietary Pattern and Risk of Invasive Breast Cancer: The Women's Health Initiative Randomized Controlled Dietary Modification Trial. JAMA. 2006;295(6):629–642. doi:10.1001/jama.295.6.629

Discussion: Comparing Study Designs

Comparing Two Studies: Think back to the two studies discussed today:

- Ecologic study: dietary fat intake and breast cancer mortality

- Randomized controlled trial: low-fat dietary intervention

Discussion questions

- What is the unit of analysis in each study?

- Which study provides stronger evidence for causality, and why?

- What kinds of conclusions are appropriate for each design?

RCTs: Strengths & Limitations

Strengths

- Randomization improves group comparability
- Strong internal validity
- Best design for evaluating causal effects of interventions

Limitations

- Often expensive and time-consuming
- May have limited generalizability to real-world populations
- Not always ethical or feasible

Project Prep 2

Revisit: Study Design & Causal Inference

Review:

Ecologic study: dietary fat & breast cancer mortality

RCT: low-fat dietary intervention

Quick question

Which design offers stronger evidence for causality?

Follow-up

What might still limit confidence in either study?

Key idea: Even strong study designs have limitations. That leads us to validity.

Challenges to the Validity of Study Designs

- Even well-designed studies can produce misleading results.

- Two central questions:

- External validity: Can the findings be generalized to other populations or settings?

- Internal validity: Is the observed association likely to reflect a true causal relationship?

- If validity is compromised, conclusions may reflect:

- Random error (chance)

- Systematic error

- Confounding

- Key idea: Strong design does not eliminate threats to validity.

External Validity

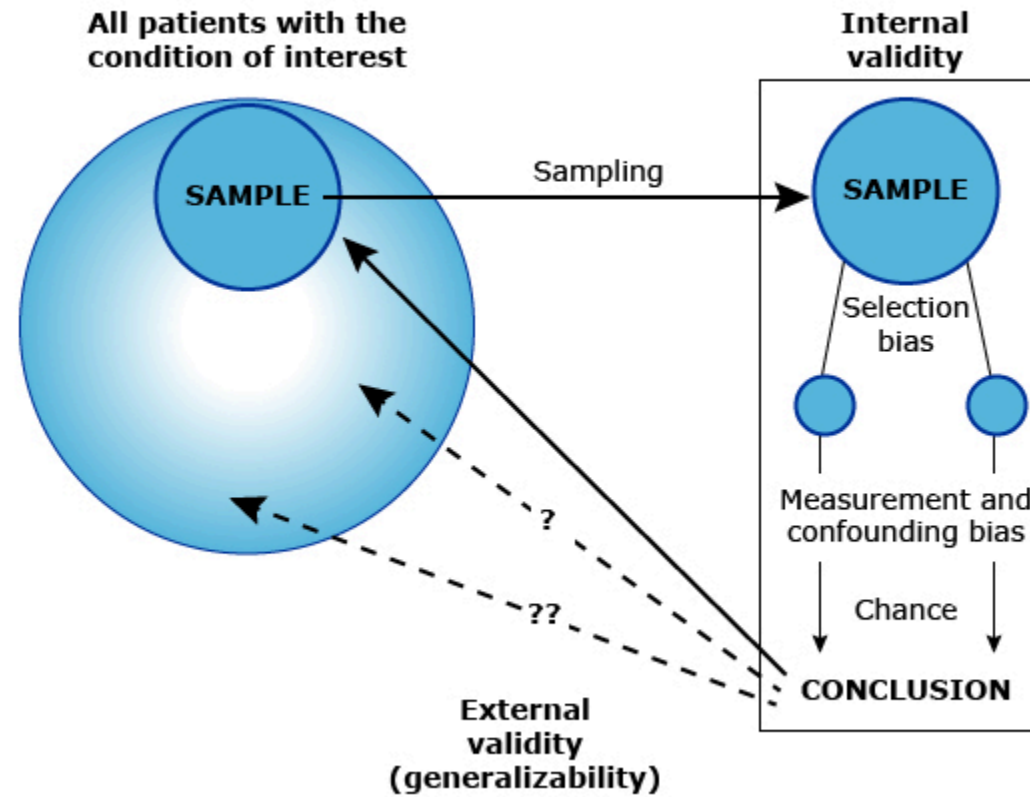
Definition: The extent to which study results can be generalized to other populations or settings

Key influences

Sampling error: random variation

Sampling bias: systematic lack of representativeness

Guiding question: Do these results apply beyond this sample?



Sampling Error

Definition: Random variation that occurs because we study a sample, not the entire population.

Key points

Occurs whenever a sample is drawn

Affects the precision of estimates

Not a design flaw — a natural result of sampling

How to reduce it: Increase sample size

Example

If the true population smoking rate is 20%,

A sample of 50 people might estimate 16% or 24% due to chance.

Key takeaway: Sampling error reflects random variation, not systematic bias.

Sampling Bias

Definition: A systematic error that occurs when the sample is not representative of the target population.

Key points

Results from how participants are selected

Affects both generalizability and potentially conclusions

Unlike sampling error, it does not disappear with larger sample size

Example

Studying hypertension using patients from a single specialty clinic may overrepresent more severe cases.

Key takeaway: Sampling bias threatens external validity because the sample does not reflect the population of interest.

Internal Validity

Random Error vs Systematic Error

- Random Error (Chance)

- Natural variability in data

- Directly affects precision (spread of estimates)

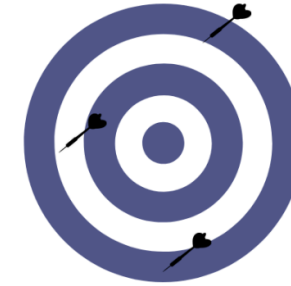
- Reduced by increasing sample size

- Systematic Error (Bias)

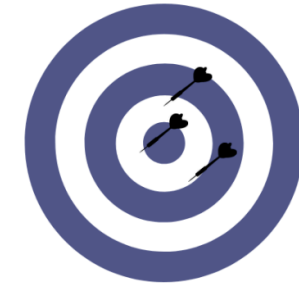
- Flaws in design, selection, or measurement

- Affects accuracy (closeness to the truth)

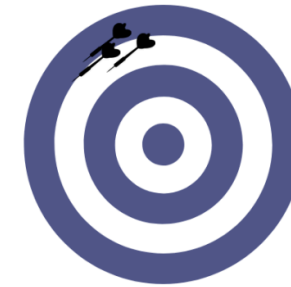
- Not corrected by larger sample size



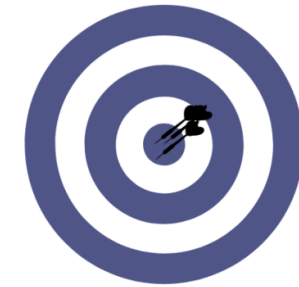
✗ low accuracy
✗ low precision



✓ high accuracy
✗ low precision



✗ low accuracy
✓ high precision



✓ high accuracy
✓ high precision

Exercise: Random or Systematic?

For each scenario, decide: Random error (chance) or Systematic error (bias)

- A random sample of 40 students is used to estimate smoking prevalence on campus. The study finds 16% prevalence, while the true campus rate is 20%.
- In a survey about alcohol use, participants are asked face-to-face by a researcher. Many participants report lower drinking levels than expected.
- In a cohort study of pesticide exposure and Parkinson's disease, more participants in the exposed group drop out during follow-up than in the unexposed group.
- Blood pressure is measured using two different devices. One device consistently gives readings 5 mmHg higher than the other.

Bias (Systematic Error)

Definition: A systematic error in the design, conduct, or measurement of a study that distorts the true association.

Key features

Shifts results in a consistent direction

Affects accuracy, not precision

Not corrected by increasing sample size

Two major categories

Selection bias

Information bias

Key idea: Bias creates a misleading estimate of the exposure–outcome relationship.

Selection Bias

Definition: A systematic error caused by differences in who enters or remains in the study.

How it occurs

- Differences in participant selection

- Differences in participation or refusal

- Differences in loss to follow-up

- Key idea: When the groups being compared differ because of the selection or retention process, the association may be distorted.

Common Types of Selection Bias

- Self-selection bias: Participation or refusal is related to exposure or outcome
- Healthy worker effect: Workers tend to be healthier than the general population
- Control selection bias: In case-control studies, controls do not represent the source population
- Loss to follow-up (attrition bias): In cohort studies, dropout differs between exposure groups
- Key idea: Selection bias occurs when group differences arise from who enters or remains in the study.

Information Bias

Definition

A systematic error caused by inaccurate or inconsistent measurement of exposure or outcome.

How it occurs

Differences in how information is reported

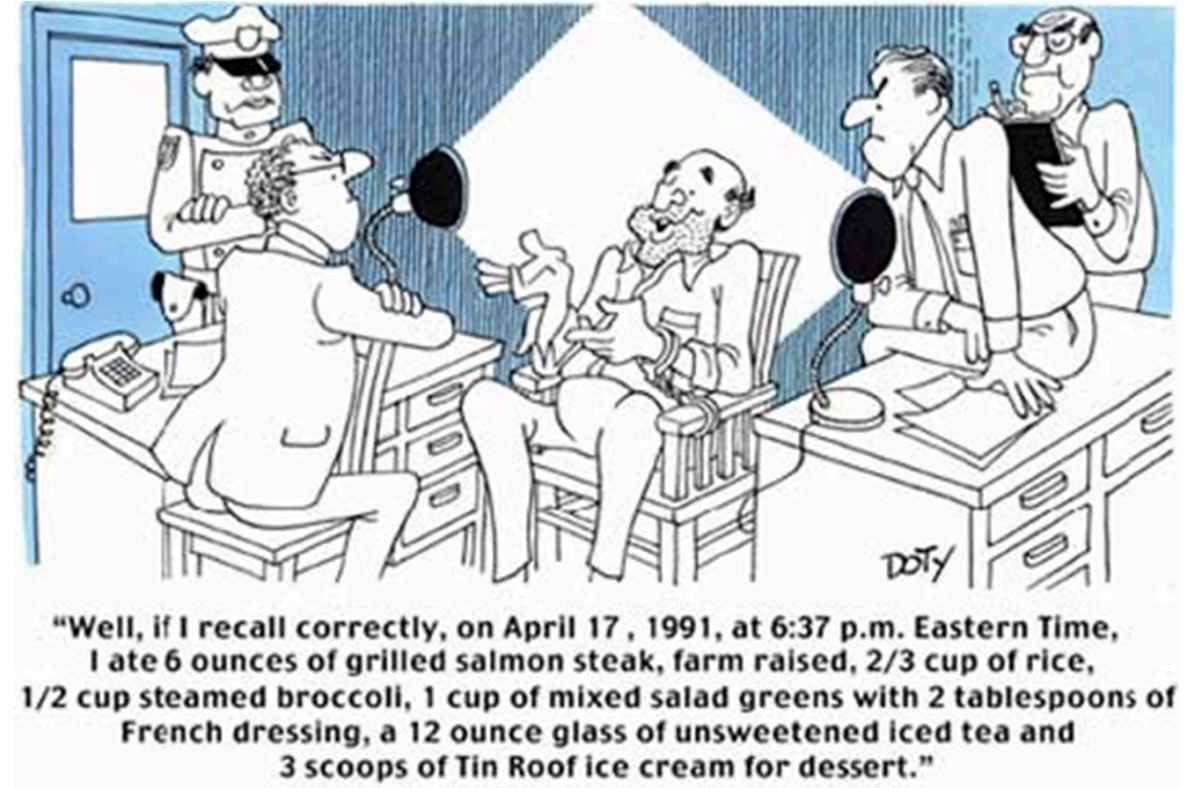
Differences in how information is collected

Differences in how information is recorded

Key idea: When exposure or outcome data differ between groups, the association may be distorted.

Types of Information Bias: Recall Bias

- Definition: A type of information bias in which participants remember past exposures differently.
- When it occurs
 - Often in case-control studies
 - Cases may recall exposures more carefully than controls
- Key idea: Differences in memory between groups can distort the exposure–outcome association.



Types of Information Bias: Researcher Bias

- Interviewer Bias
 - Occurs during data collection
 - Interviewer knowledge influences how questions are asked
 - Can lead to different responses between groups
- Observer Bias
 - Occurs during outcome assessment
 - Observer expectations influence how outcomes are recorded
 - More likely when blinding is absent
- Key idea: When researchers influence data collection or measurement, results may be systematically distorted.



Types of Information Bias: Social Desirability

- Definition: participants provide responses that are viewed as socially acceptable rather than accurate.

- When it occurs

- Sensitive behaviors (e.g., smoking, alcohol use, drug use)

- Self-reported health behaviors or lifestyle factors

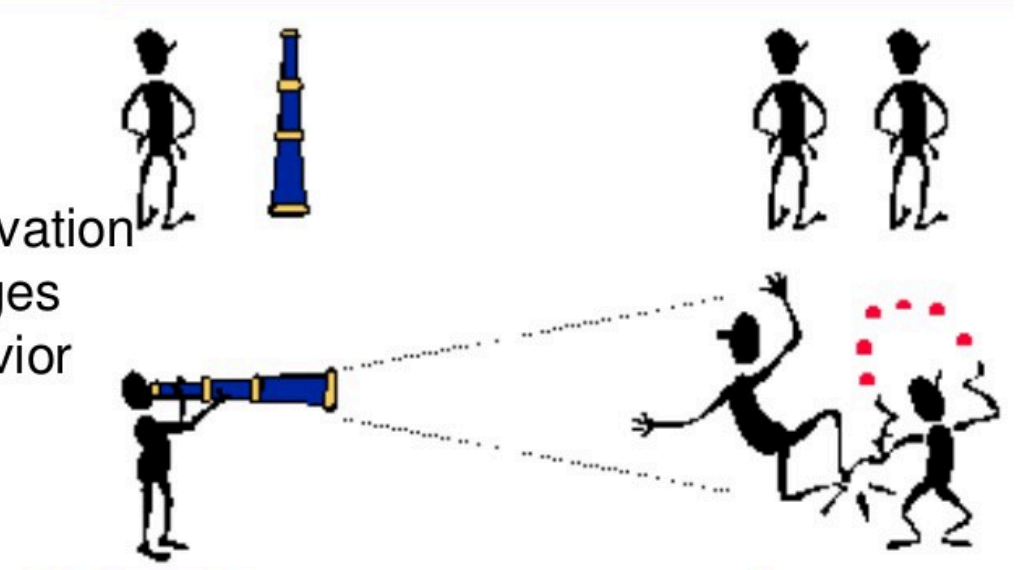
- Key idea: When participants modify responses to appear favorable, exposure or outcome data may be systematically distorted.



Types of Information Bias: Hawthorne effect

- Definition: participants change behavior because they know they are being observed.
- When it occurs
 - During clinical trials or behavioral studies
 - When participants are monitored or aware of measurement
- Key idea: when behavior changes due to observation, the measured exposure or outcome may not reflect usual behavior.

Observation
Changes
Behavior



Exercise: Identify the Type of Information Bias

For each scenario, identify the most likely type of information bias and explain why.

- In a case-control study, parents of sick children recall pesticide exposure during pregnancy more carefully than parents of healthy children.
- Participants report lower alcohol use in face-to-face interviews than expected from sales data.
- In a clinical trial, outcome assessors know which participants received the treatment and rate them more favorably.
- Interviewers ask more detailed exposure questions to lung cancer cases than to controls.
- Participants increase exercise levels during a study because they know their activity is being monitored.

Reducing Selection and Information Bias

Reducing Selection Bias

- Use appropriate comparison groups

- Apply clear and consistent inclusion criteria

- Minimize loss to follow-up

- Ensure recruitment is not related to both exposure and outcome

Reducing Information Bias

- Use standardized definitions and procedures

- Apply blinding when possible

- Collect data consistently across groups

- Use validated measurement tools

- Key idea: Careful design and consistent measurement strengthen internal validity.

Confounding

Definition: a distortion of the exposure–outcome association caused by a third variable.

How to identify a confounder

Associated with the exposure

Independently associated with the outcome

Not caused by the exposure

Example

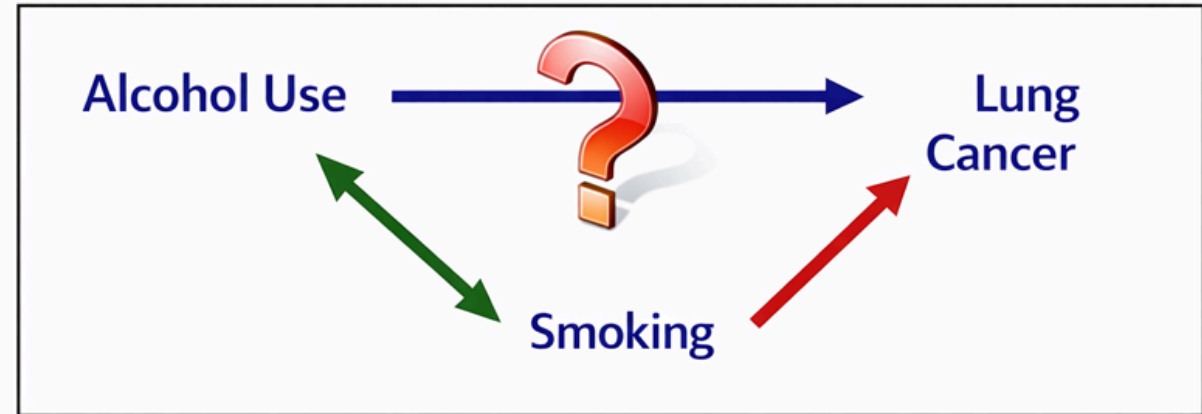
Smoking may confound the association between alcohol use and lung cancer.

Key idea: If a third variable meets these criteria, the observed association may not reflect a true causal relationship.

Scenario

- Studying the effect of Alcohol Use (exposure) on Lung Cancer (outcome).
 - Confounder: Smoking
 - Associated with alcohol use
 - Independently associated with lung cancer
 - Not caused by alcohol use
 - Effect
 - The observed association between alcohol and lung cancer may be distorted because smoking is related to both.

Example of Confounding



- ✓ Associated with the exposure (Alcohol use)
- ✓ Independently associated with the outcome (Lung cancer)
- ✓ Not on the causal pathway

Exercise: Could This Be Confounding?

For each scenario, decide whether the association could be explained by confounding. Identify a possible confounder and explain why.

- Scenario 1: A study finds that people who carry lighters have a higher risk of lung cancer.

- What third variable might explain this association?

- Does it meet the criteria for a confounder?

- Scenario 2: A study finds that children who sleep with the light on are more likely to develop myopia (nearsightedness).

- What third variable might explain this association?

- Does it meet the criteria for a confounder?

Addressing Confounding

- Confounding cannot be ignored when interpreting results.
- Randomization helps balance confounders between groups
- In observational studies, researchers must identify and measure potential confounders
- Results should be interpreted cautiously if important confounders are not accounted for
- Key idea: If confounding is not addressed, an observed association may not reflect a true causal relationship.

Wrap-Up

Causal inference depends on:

Study design

Observational vs experimental

Ecologic vs RCT

Validity

External validity

Internal validity

Threats to internal validity

Chance

Selection bias

Information bias

Confounding

Key message: Strong conclusions require both good design and strong validity.

Final Epi Presentation
Get started!